

27 D

$$QV = E_k = \frac{p^2}{2m}$$

$$p = \sqrt{2mQV} = \frac{h}{\lambda}$$

$$\lambda = \frac{h}{\sqrt{2mQV}}$$

$$\text{Hence } \lambda \propto \frac{1}{\sqrt{mQ}}$$

Shortest wavelength corresponds to greatest product of mass and charge.

$$\lambda \propto \frac{1}{\sqrt{4(1.66 \times 10^{-27})(2 \times 1.60 \times 10^{-19})}} = 2.17 \times 10^{22}$$

$$\text{Option A: } \lambda \propto \frac{1}{\sqrt{(1.67 \times 10^{-27})(1.60 \times 10^{-19})}} = 6.11 \times 10^{22}$$

$$\text{Option B: } \lambda \propto \frac{1}{\sqrt{(9.11 \times 10^{-31})(1.60 \times 10^{-19})}} = 2.62 \times 10^{24}$$

$$\text{Option C: } \lambda \propto \frac{1}{\sqrt{2(1.66 \times 10^{-27})(1.60 \times 10^{-19})}} = 4.33 \times 10^{22}$$

28 C Three spectral lines from Y means three energy levels for Y